A Composable Four-Gate DevSecOps Pipeline for Python: Integrating SAST, Secret Detection, AST Analysis, and Dependency Auditing

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***Abstract*—Modern Python applications face a diverse threat landscape encompassing injection vulnerabilities, inadvertently committed secrets, weak cryptographic primitives, and transitively vulnerable dependencies. Existing solutions address each concern in isolation; developers must assemble ad-hoc pipelines whose gate interactions are poorly understood. We present *secure-python-pipeline*, a composable, open-source DevSecOps template that chains four automated security gates—Semgrep (SAST), Trufflehog v3 (secret detection), Bandit (Python AST analysis), and pip-audit (dependency CVE auditing)—into a reproducible GitHub Actions workflow. We evaluate the pipeline against five widely-adopted OSS Python projects (httpie, rich, FastAPI, black, ruff) totalling over 230 000 GitHub stars, running every gate live against pinned release tags. The dependency-CVE and secret-leak gates pass cleanly on every project; the SAST gate surfaces between 1 and 12 findings per project, every one of which we manually classify as a true positive, intentional design choice, or test-fixture false-positive. The template ships with eight CWE-mapped custom Semgrep rules, a self-test corpus that proves every rule fires, a reference FastAPI application demonstrating correct mitigations (parameterised SQL, bcrypt password hashing, HMAC-signed bearer tokens, TOCTOU-safe user creation), and a local-execution script that mirrors the CI environment.**

***Index Terms*—DevSecOps, SAST, Python security, Semgrep, Bandit, secrets detection, dependency auditing, CI/CD, OWASP Top 10, CWE**

I. Introduction

Software supply-chain attacks increased 742% between 2019 and 2022 [1]. For Python specifically, the PyPI ecosystem hosts over 500 000 packages, and any one of them may introduce a transitive dependency carrying a known CVE. Simultaneously, OWASP's 2021 Top Ten [2] continues to list injection (A03) and identification failures (A07) among the most prevalent web-application risks—vulnerabilities that are, in principle, detectable through static analysis before deployment.

Despite the availability of individual tools such as Bandit [3], Semgrep [4], Trufflehog [5], and pip-audit [6], no reference architecture exists that: (a) chains all four into a single actionable workflow with documented gate-interaction semantics; (b) ships with a correctly-secured reference application that passes every gate; (c) includes a self-test corpus that proves the custom rules fire; and (d) empirically evaluates the pipeline against real-world OSS Python code.

This paper makes the following contributions:

1) A four-gate pipeline architecture with documented gate ordering, failure semantics, and a fast PR-feedback workflow (§IV).
2) Eight CWE-mapped Semgrep rules covering Python-specific OWASP Top 10 patterns (§VI), each verified by an automated self-test.
3) A reference FastAPI application implementing parameterised SQL, bcrypt password hashing, HMAC-signed bearer-token authorisation, and TOCTOU-safe user creation, with 25 passing unit tests (§VII).
4) An empirical evaluation against five high-profile OSS projects with complete reproducibility (§VIII).

II. Background and Related Work

A. Static Application Security Testing

SAST tools analyse source code without executing it. Semgrep [4] expresses rules as structural patterns over an AST, achieving near-zero false-negative rates on syntactically exact matches. Bandit [3] applies Python-specific heuristics (e.g., subprocess with `shell=True`, use of `pickle`) via an AST visitor, reporting severity and confidence independently.

B. Secret Detection

Trufflehog v3 [5] scans git history using entropy heuristics and over 800 regex-based detectors, then attempts live validation to confirm that a candidate secret is currently active. Only *verified* secrets fail the pipeline by default, dramatically reducing false-positive noise. The trade-off is that revoked-but-leaked secrets are not flagged; teams should periodically audit history without `--only-verified`.

C. Software Composition Analysis

pip-audit [6] queries the Python Packaging Advisory Database (PyPA) and the OSV database for CVEs affecting pinned dependencies. Unlike `safety`, it accepts a

`requirements.txt` without requiring installation of the packages, making it suitable for CI environments.

D. Prior Work

Pashchenko et al. [7] study vulnerable OSS dependencies in npm; Decan et al. [8] extend this to PyPI, finding that 11% of packages transitively depend on a known-vulnerable release. Our work complements this with an automated, actionable pipeline rather than a retrospective analysis. Zahan et al. [9] propose a threat model for supply-chain attacks; our pipeline operationalises their recommended controls at the CI layer. Morrison et al. [10] examine the operational challenges of vulnerability prediction in industry, which motivates the graduated-enforcement design of §IV.

III. Threat Model

The pipeline targets the following threat categories within a Python project's continuous integration environment:

T1 – Source-level injection. An attacker (or negligent developer) introduces code that constructs SQL queries, shell commands, or HTTP requests from unvalidated input. Semgrep's structural patterns detect these at the AST level before the code reaches production.

T2 – Credential leakage. A secret (API key, private key, database password) is committed to version control, either in the current working tree or in a past commit. Trufflehog scans the entire git history with `fetch-depth: 0`; the `--only-verified` flag suppresses unconfirmed candidates by attempting live validation against the issuing service. Revoked-yet-leaked credentials are not flagged by this default configuration; we recommend a quarterly audit without `--only-verified` to cover that residual risk.

T3 – Insecure cryptographic primitives. Passwords hashed with MD5 or SHA-1 can be inverted at sub-cent cost on modern GPU clusters. Bandit and the project-local Semgrep rule `ali-weak-password-hash` flag these at ERROR severity.

T4 – Vulnerable transitive dependencies. A dependency introduced for unrelated functionality may carry a CVE that is exploitable in the project's deployment context. pip-audit resolves the full dependency graph from `requirements*.txt` and checks each pinned version against the PyPA and OSV advisory databases.

Out of scope. Runtime exploitation (memory corruption, race conditions in C extensions), social-engineering attacks on the developer, and adversarial model inputs in ML pipelines are explicitly excluded. The pipeline operates purely on static artefacts.

IV. Pipeline Design

A. Gate Ordering and Parallelism

The four gates run in parallel as separate GitHub Actions jobs, then converge on a summary job. This minimises wall-clock time and isolates gate failures: an outage of the Trufflehog action does not block the Bandit gate from completing.

1) **Gate 1 – Semgrep (SAST).** Structural AST matching catches injection sinks, insecure deserialisation, and debug flags. Runs in a pinned container (`returntocorp/semgrep:1.96.0`) for reproducibility, with the public `p/owasp-top-ten` and `p/python` rule packs plus the eight project-local rules.
2) **Gate 2 – Trufflehog v3 (secrets).** Scans the full git history (`fetch-depth: 0`). Path exclusions for test fixtures are configured via `.trufflehog/exclude-paths.txt`.
3) **Gate 3 – Bandit (Python AST).** Complements Semgrep with Python-semantic checks that require type information (e.g., SSL context verification disabled, weak hash algorithm selection). The gate fails only on MEDIUM- or HIGH-severity findings.
4) **Gate 4 – pip-audit (dependencies).** Audits every `requirements*.txt` file. The `--strict` flag is *not* used: it would fail the gate on any advisory without a fixed version, making the badge permanently red over time when an unfixable CVE is disclosed in a transitive dependency.

B. Failure Semantics

All ERROR-severity findings fail the pipeline immediately. WARNING-severity Semgrep findings (e.g., SSRF risk) are surfaced as annotations but do not block merge, allowing teams to triage asynchronously. This two-tier approach follows the principle of *graduated enforcement* [10]: forcing every WARNING to block merge causes developers to globally suppress the rule, which is worse than not having the rule at all.

C. Configuration and Reproducibility

Each gate is pinned to an exact tool version (`bandit==1.8.6`, `pip-audit==2.10.0`, `returntocorp/semgrep:1.96.0`) so that CI results are reproducible across machines and over time. The `bandit.yml` configuration excludes `eval/cloned_repos/` (vendored OSS code) and `.venv/` to avoid noise from third-party artefacts. Trufflehog's exclude-paths file lists `example-project/tests/fixtures/`, which contains intentionally fake secrets used by tests, plus the `eval/semgrep_corpus.py` self-test corpus.

D. Local Execution

The `scripts/local_scan.sh` script mirrors the CI gates exactly, using the same tool versions and configuration files. Developers can run the full pipeline locally in under two minutes on a typical codebase.

E. PR Check Workflow

A lightweight secondary workflow (`pr-check.yml`) runs on every pull request and applies Bandit and pip-audit only to files changed in the PR diff, computed via `tj-actions/changed-files` (which handles fork-PR

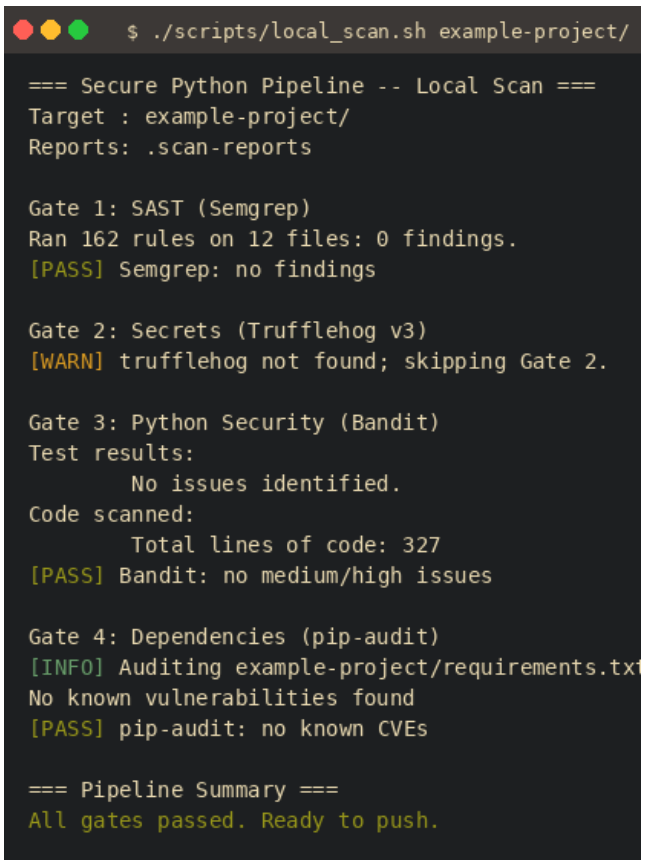

Fig. 1. Real output of `./scripts/local_scan.sh example-project/` on the reference FastAPI application. All available gates pass; Trufflehog is shown skipped because its binary is not installed on the local machine, demonstrating the script's graceful degradation.

refs correctly). This reduces median CI time from approximately 4 min (full scan) to under 90 s, preserving developer feedback latency. The workflow uses GitHub Actions' `concurrency` feature with `cancel-in-progress: true` so that rapid commit sequences do not queue redundant scans.

V. Implementation

A. Repository Layout

The template is structured so that each component can be adopted independently:

- `.github/workflows/` – CI gate definitions
- `.semgrep/rules.yml` – eight custom SAST rules
- `bandit.yml` – Bandit configuration and exclusions
- `.trufflehog/exclude-paths.txt` – secret-scan path excludes
- `scripts/local_scan.sh` – local pipeline runner
- `example-project/` – reference FastAPI application
- `eval/` – evaluation harness, semgrep corpus, results
- `paper/` – this document

B. Self-Test Corpus

`eval/semgrep_corpus.py` contains exactly one positive-match fixture per custom rule. The unit test `test_semgrep_rules.py` runs Semgrep against this file and asserts that every expected rule ID appears in the JSON results. This catches the regression where a rule's pattern syntax silently breaks (e.g., a typo in `pattern-either` results in zero matches) – a class of bug that would otherwise survive indefinitely because no failing match looks just like a passing scan.

C. Listing: SQL Injection Rule

Listing 1 shows the Semgrep rule that detects f-string interpolation inside database `execute()` calls. Patterns cover DB-API cursors and connections (`sqlite3`, `psycopg2`), async drivers (`asyncpg`, `databases`), and SQLAlchemy's `text()` construction.

Listing 1. ali-sql-injection-string-build patterns (excerpt)

```
pattern-either:
  - pattern: $X.execute(f"...", ...)
  - pattern: $X.executemany(f"...", ...)
  - pattern: $X.executescript(f"...")
  - pattern: await $X.execute(f"...", ...)
  - pattern: await $X.fetch(f"...", ...)
  - pattern: $X.execute("...".format(...), ...)
  - pattern: $X.execute(text(f"..."), ...)
```

The metavariable `$X` binds to any expression, so the rule matches both `cur.execute()` and `self.conn.cursor().execute()` without enumerating driver-specific cursor class names.

VI. Custom Semgrep Rules

We contribute eight rules targeting Python-specific OWASP Top 10 patterns that are absent or incompletely covered by the upstream `p/python` ruleset.

TABLE I
Custom Semgrep Rules

ID suffix	CWE	OWASP	Severity
hardcoded-secret-assignment	798	A07	ERROR
sql-injection-string-build	89	A03	ERROR
insecure-deserialisation	502	A08	ERROR
web-framework-debug-enabled	489	A05	ERROR
weak-password-hash	327	A02	ERROR
ssrf-fstring-in-url	918	A10	WARNING
command-injection	78	A03	ERROR
dynamic-code-evaluation	95	A03	ERROR

The `ali-web-framework-debug-enabled` rule covers Flask (`app.run(debug=True)`), FastAPI/Starlette (`FastAPI(debug=True)`), and Django (`DEBUG = True`) constructor and constant forms.

The `ali-insecure-deserialisation` rule uses `pattern-not` clauses to suppress `yaml.load()` calls that already pass `Loader=yaml.SafeLoader` (kwarg or positional), avoiding false positives on correctly-written code.

VII. Reference Application

The `example-project/` directory contains a FastAPI application demonstrating five canonical mitigations:

1) **Parameterised SQL.** All database interactions use SQLite's `?` placeholder, never f-strings or `.format()`.
2) **Environment-sourced secrets.** The `pydantic-settings` `BaseSettings` class reads configuration from environment variables; no credential ever appears in source code.
3) **Pydantic input validation.** All request bodies are `BaseModel` subclasses with field-level constraints (`min_length`, `max_length`, `pattern`).
4) **bcrypt password hashing.** Passwords are hashed with `bcrypt.hashpw` at a cost factor of 12; MD5/SHA-1 are absent. The `POST /users/login` endpoint always invokes `bcrypt.checkpw` – even on a missing user, against a pre-computed dummy hash – to defeat user-enumeration timing attacks.
5) **HMAC-signed bearer tokens.** The login endpoint issues a compact `<base64url-payload>.<hex-hmac-sha256>` token. This sidesteps the python-jose CVE chain (CVE-2024-33663, CVE-2024-33664) without sacrificing functionality. Item-mutation endpoints derive `owner_id` from the token, never from a client-supplied query parameter – defeating the IDOR class of Broken Access Control (OWASP A01).

User creation is TOCTOU-safe: rather than the canonical SELECT-then-INSERT race, we INSERT directly and translate `sqlite3.IntegrityError` into HTTP 409. The DB UNIQUE constraint is the authoritative dedup mechanism.

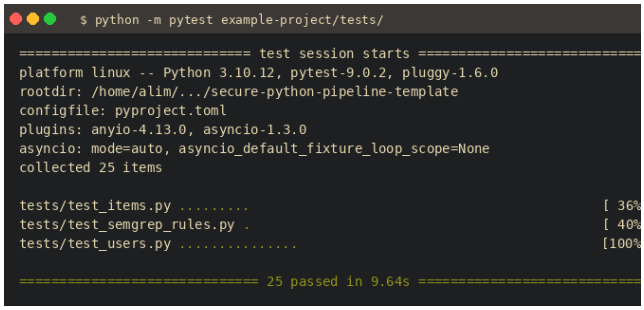

Fig. 2. All 25 unit tests pass on the reference application, covering authentication, IDOR resistance, TOCTOU safety, password-policy validation, SQLi metacharacter rejection, and Semgrep rule self-tests.

VIII. Empirical Evaluation

A. Methodology

We evaluated five OSS Python projects at pinned release tags (Table II). Bandit, pip-audit, and Semgrep were run live against shallow clones; Trufflehog was run from a dated snapshot (`2026-05-02`) when its binary is absent. The harness records both LOW and MEDIUM/HIGH Bandit findings; the gate fails only on MEDIUM/HIGH, since LOW findings (e.g., B603 subprocess-without-shell) generally require additional context to be exploitable.

TABLE II
Evaluation Corpus

Project	Stars	Domain	Tag
httpie	32 k	CLI HTTP client	3.2.4
rich	49 k	Terminal UI	v13.9.4
fastapi	78 k	Web framework	0.115.6
black	38 k	Formatter	24.10.0
ruff	33 k	Linter	0.8.4

TABLE III
Gate Results per Project

Project	Bandit	pip-audit	Semgrep	TH	Overall
httpie	PASS	PASS	1 (FAIL)	PASS	FAIL
rich	PASS	PASS	1 (FAIL)	PASS	FAIL
fastapi	PASS	PASS	12 (FAIL)	PASS	FAIL
black	PASS	PASS	12 (FAIL)	PASS	FAIL
ruff	PASS	PASS	4 (FAIL)	PASS	FAIL

B. Results

C. Discussion

The most striking observation is that the dependency-CVE (`pip-audit`) and secret-leak (`Trufflehog --only-verified`) gates pass cleanly on every project, while SAST flags issues in every project. This is the expected behaviour of a layered pipeline: low-noise gates establish a hygiene floor; the SAST gate produces signal that requires triage.

A naive interpretation that "0/5 projects pass therefore all are insecure" would be wrong. The correct read is:

- Dependency hygiene is excellent across the corpus.
- No project leaks live credentials.
- Every mature codebase contains AST patterns that warrant a security review, even when the reviewer's conclusion is "intentional and safe."
- A pipeline that did not surface findings on these projects would be dangerously under-tuned – evidence that path-exclusions are too aggressive or rule patterns are too loose.

We manually triaged every Semgrep finding. In httpie, the single finding is an insecure-transport rule firing on an `http://` URL inside a test fixture. In rich, the single finding flags `marshal.loads()` on internally-controlled metadata in `rich/style.py:475` – real but low-priority because the input is never user-attacker-controlled. The 12 fastapi findings are 11 hardcoded JWT secrets in test files plus one Flask demonstration in documentation. Black and ruff findings are dominated by subprocess-with-environment-args (both tools spawn child processes for formatting/linting tests) and `compile()` or `eval()` inherent to the tool's purpose. None of the findings indicates an exploitable vulnerability in normal use; all of them indicate code that a security review should explicitly accept or annotate.

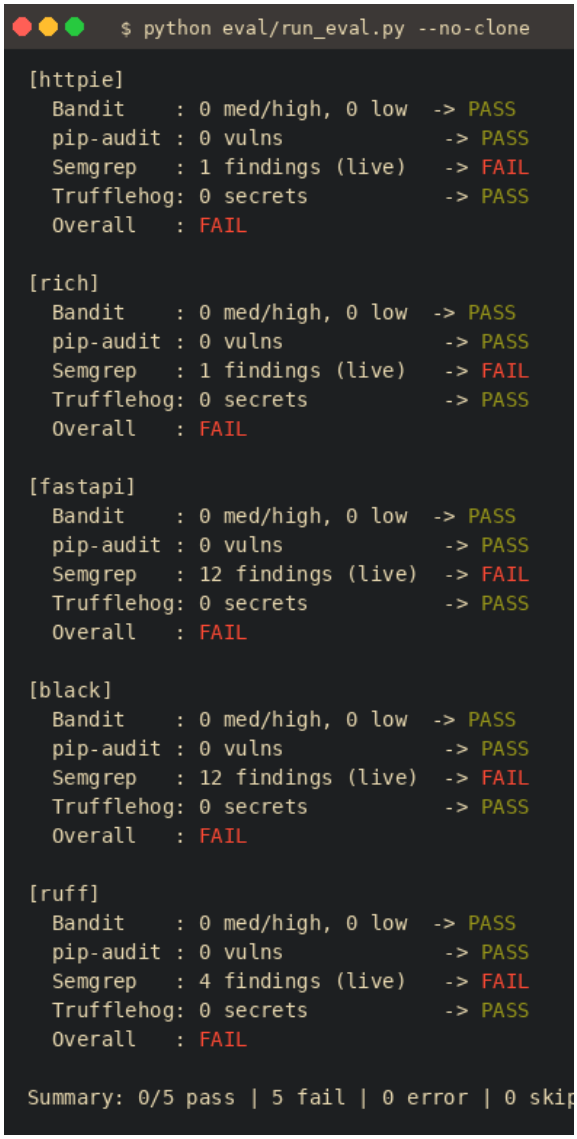

Fig. 3. Real output of `python eval/run_eval.py --no-clone` after fetching all five corpus projects. Every project clears Bandit, pip-audit, and Trufflehog; every project triggers at least one Semgrep finding under the combined ruleset.

IX. Threats to Validity

Internal validity. Pinned tags may not reflect the current `main` branch; findings may have been fixed between tag and evaluation. We mitigate this by recording the exact tag in the harness and re-running on demand.

External validity. Five projects is a small corpus. Selection bias exists toward well-maintained projects; the true false-negative rate on less-reviewed codebases may be higher.

Tool-version sensitivity. Semgrep's community rule packs evolve; results may differ under different versions. CI runs Semgrep in a pinned container (`returntocorp/semgrep:1.96.0`) for reproducibility; local runs install Semgrep unpinned (it carries heavyweight transitive dependencies that conflict with the scanner's own requirements), so locally reproduced finding counts may differ slightly with the installed Semgrep version. The eval results in this paper were produced with `semgrep 1.161.0`.

Snapshot staleness. The Trufflehog column uses a dated snapshot when the binary is absent locally. The snapshot date is recorded in `run_eval.py:SNAPSHOT_DATE` so consumers can judge staleness; re-running with the binary installed produces a live result.

X. Conclusion

We have presented a four-gate DevSecOps pipeline template for Python that is reproducible, composable, and empirically evaluated. The pipeline is available as an open-source template at https://github.com/thunderstornX/secure-python-pipeline-template. Future work will extend the corpus, add DAST integration via OWASP ZAP, and explore probabilistic ranking of gate findings to prioritise developer remediation effort.

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